Umberto Domenico Ciccia

Site Reliability Engineer, Software Oriented | Cloud Infrastructure | Distributed Systems & Automation

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Experience

Amazon Web Services (AWS), Dublin, Ireland, Nov 2025 – present
System Development Engineer

- Diagnosed and corrected a DNS monitoring misconfiguration in a three-AZ setup, where same-AZ availability checks failed due to NAT hairpin behavior, generating false-positive SEV-2 pages; eliminated the recurring alert noise, reducing the total number of SEV-2 alerts from 84 to zero over six months.
- Implemented DNSSEC support for Amazon Route 53 in AWS European Sovereign Cloud three weeks ahead of schedule, delivering API and infrastructure changes, DNS record change-propagation updates, regional allowlisting, monitoring, console integration, pipeline updates, and automated host patching to unblock a strategic customer migration.
- Delivered CIDR-based DNS routing four months ahead of schedule, implementing infrastructure, API, and console changes to enable customer-configurable IP-based DNS routing.
- Refactored 184 Route 53 infrastructure pipelines, increasing the proportion with artifacts less than 40 days old from 80% to 98% in eusc-de-east-1.
- Supported production reliability for Route 53, AWS WAF, and AWS Shield across AWS European Sovereign Cloud through on-call incident response, monitoring, root-cause analysis, and corrective actions.
- Reviewed peer code changes and validated software and infrastructure changes through unit, integration, contract, end-to-end, and canary testing; documented incident findings and troubleshooting procedures in postmortems and operational runbooks.

NTT Data, Cosenza, Italy, Jan 2025 – Oct 2025
Cloud Software Engineer

- Replaced fragile, manual Bash deployments with Jenkins CI/CD pipelines for infrastructure and backend applications, automating Terraform provisioning, container image builds and publication to Amazon ECR, and deployments to Amazon EKS; introduced manual approval gates for infrastructure changes and automated application rollback based on API error rates and p50, p90, and p99 latency.
- Configured AWS Database Migration Service (DMS) for both initial data loading and ongoing change replication during migration from on-premises MariaDB to Amazon Aurora, supporting the transition of application data to AWS.
- Built GCP data ingestion pipelines processing 10k+ daily records, improving data availability and release reliability using Dataflow, Compute Engine, and Cloud Storage for Coop Italia
- Built an AI-powered Quarkus agent using the Vertex AI SDK to analyze user-defined multi-cloud architectures and recommend optimized configurations, reducing estimated infrastructure costs by 30% versus the original designs on the NTT KUMO platform.

Vision One, Recanati, Italy, July 2024 – Dec 2024
Software Engineer

- Developed an organization-wide on-call scheduling system using .NET and Blazor, implementing rule-based shift assignment, availability validation, and automatic shift reassignment to resolve scheduling conflicts; integrated database and internal API interactions across backend, data, and frontend layers.
- Improved scheduling and management efficiency by 80% by automating scheduling workflows.
- Removed 60% of unused code and introduced xUnit tests achieving 80% test coverage.

Caliò Informatica, Cosenza, Italy, Jan 2024 – July 2024
Software Engineer

- Developed frontend and backend components for an AI-powered visual workflow builder, enabling users to create and manage workflows composed of connected steps integrating AI models and external APIs/services.
- Optimized API interactions in an AI-powered visual workflow builder by eliminating redundant requests and combining calls, achieving an 80% reduction in API costs and a 60% performance improvement, validated through pre- and post-optimization profiling.

Skills

SRE & Reliability Engineering: SLIs/SLOs, Error Budgets, Incident Management, Root Cause Analysis (RCA), On-Call Operations, Capacity Planning, Performance Engineering, Resilience & Disaster Recovery

Cloud & Infrastructure: AWS, GCP, Azure, Linux, TCP/IP, HTTP/TLS, DNS, Kubernetes, Docker, Helm, Terraform, Pulumi, Crossplane, AWS CDK, Ansible

Observability: Prometheus, Grafana, Monitoring, Alerting, Logging, Distributed Tracing

CI/CD & Automation: Jenkins, Argo CD, Git, Bash, Python, PowerShell, Safe Deployments, Change Management, Infrastructure Automation

Programming Languages, Frameworks & Databases: Java, TypeScript, JavaScript, C#, SQL, C++, Quarkus, Spring Boot, .NET, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, DynamoDB

Languages: Italian (Native), English (Professional)

Education

Università della Calabria, M.Sc. in Artificial Intelligence and Machine Learning – GPA: 4.0/4.0

Università della Calabria, B.Sc. in Computer Engineering – GPA: 4.0/4.0